

MICHAEL THROOLIN

✉ michael.throolin@utah.edu 🌐 mthroolin.github.io in www.linkedin.com/in/mthroolin

SUMMARY

PhD student in Population Health Sciences (emphasis in Biostatistics) at the University of Utah, currently funded through a Research Assistantship and a TL1 Predoctoral Fellowship (Mountain States ASCEND, NIH CTSA). Interests in causal inference, machine learning, and missing data analysis. Fluent in Portuguese and conversational in Spanish.

EDUCATION

PhD Student - Population Health Sciences (Biostatistics) University of Utah Awarded Mountain States ASCEND TL1 Predoctoral Fellowship (NIH CTSA)	Expected Completion 2027
MS Statistics Montana State University Scholarships: Kenneth J. Tiahrt and John L. Magaret Math Scholarships	May 2023
BS Mathematics, Summa Cum Laude Brigham Young University - Hawaii Scholarships: Math Department, Recruitment, and Merit Minor in Computer Science	June 2016

EXPERIENCE

Research Assistant University of Utah, Salt Lake City, UT	2023 - Present
• Providing statistical support to papers published by the Department of Population Health Sciences. • Atrial fibrillation research and kidney disease research with the VA Medical Center.	
Graduate Teaching Assistant Montana State University, Bozeman MT	2021 - 2023
• Course Assistant for Methods of Data Analysis II (STAT 412/512). Topics included: experimental design, multi-factor ANOVA, repeated measures, logistic regression, Poisson log-linear regression, and introductions to multi-variate and time series analyses. • Co-instructor for an introductory course in statistics (STAT 216). Topics included: Traditional and resistant estimators of location and spread, fundamentals of inference using randomization and classical methods, confidence intervals, and tests of hypotheses.	
Instructor Johns Hopkins Center for Talented Youth, Baltimore, MD	Summer 2022
• Developed curriculum and instructed a summer course entitled "Paradoxes and Infinities" to a group of middle school and high school students. Topics included set theory, topology, computer theory, game theory, calculus, moral paradoxes, vagueness paradoxes, self referential paradoxes, and several mathematical paradoxes.	

PUBLICATIONS

2025

- Derington CG, Sarwal A, Wei G, Hartsell SE, **Throolin M**, Singh R, Nevers MR, Drakos SG, Greene T, Shen J, Beddhu S. Liraglutide vs Semaglutide vs Dulaglutide in Veterans With Type 2 Diabetes. *JAMA Network Open*. 2025;8(10):e2537297. doi: 10.1001/jamanetworkopen.2025.37297

- Jacobson BT, DeWit-Dibbert J, Zanca L, Sonar S, Hardy C, **Throolin M**, Brewster P, Andujo K, Jones K, Sago J, Smith S, Bowen L, Bimczok D. Pathogen delivery route impacts disease severity in experimental *Mycoplasma ovipneumoniae* infection of domestic lambs. *Vet Res* 56, 10 (2025). doi: 10.1186/s13567-024-01439-y

2024

- Jacobson BT, DeWit-Dibbert J, Selong ET, Quirk M, **Throolin M**, Corona C, Sonar S, Zanca L, Schwarz ER, Bimczok D. Innovative Methodology for Antimicrobial Susceptibility Determination in *Mycoplasma* Biofilms. *Microorganisms*. 2024; 12(12):2650. doi: 10.3390/microorganisms12122650

SELECTED COURSEWORK

Advanced Topics: Clinical Trials, Causal Inference, Bayesian Analysis, Advanced Statistical Inference, EHR Data Analysis, High Dimensional Data Analysis, Survival Data Analysis, Epidemiology Theory and Methods

Statistical Methods: Multilevel Models, Mixed Effects Models, Time Series Analysis, Multivariate Analysis, Experimental Design, Linear Models/Regression Analysis

Computing: Advanced Computing in R, Computational Theory, Algorithms and Complexity

Mathematics: Real Analysis, Abstract Algebra, Numerical Methods, Differential Equations, Multivariate Calculus